Shanghai Pinghe School, High School Dept. AP Program

Name:_____ Class:_____

First Monthly Exam （Version A）

Autumn Semester 2024-2025

G11 Regular Biology

Duration: 50 minutes

INSTRUCTION

- ✧ **Do not open the exam paper until you are instructed to do so.**
- ✧ **Answer all questions on the paper provided.**
- ✧ **The full mark for this exam is 100 marks.**
- ✧ **Do not hand- in the paper before the end of the exam.**
- ✧ **The total exam paper contains 6 Pages**

Part I: Multi-Choice Questions 20*3=60 marks

1. Which level of analysis in neuroscience may ask the question of how many different types of neurons exist in the brain of a newly discovered species, and how they differ in function?

A. cellular B. systems C. behavioral D. cognitive

2. Which of the following is a loss of feeling and movement caused by traumatic damage to the spinal cord?

A. autism B. depression C. spinal paralysis D. stroke

3. Pierre Flourens performed experimental ablations of the cerebellum in pigeons and determined that the structure plays a role in the coordination of movement. That he later repeated this work in rabbits could be considered an example of

A. observation. B. replication. C. interpretation. D. verification.

4. During the Renaissance, this anatomist is primarily credited for adding more detail to the structure of the brain.

A. Vesalius B. Galen C. Descartes D. Hippocrates

5. Which of the following disorders emerges in early childhood and is characterized by impairments in communication and social interactions, and restricted and repetitive behaviors?

A. autism B. depression C. epilepsy D. stroke

6. Which of the following is a severe psychotic illness characterized by delusions, hallucinations, and bizarre behavior?

A. autism B. cerebral palsy C. epilepsy D. schizophrenia

7. Bumps on the surface of the brain are known as

A. sulci. B. gyri. C. lobes. D. gray matter.

8. Which of the following nineteenth century scientists showed that applying small electrical currents to a certain region of the surface of the brain of a dog could elicit discrete movements?

A. Fritsch and Hitzig B. Ferrier C. Munk D. Broca

9. The thin projections that extend from the cell body of neurons are known as

A. ventricles. B. neurites. C. meninges. D. glia.

10. What type of cell senses changes in the environment, communicates these changes to other cells of that type, and commands the body's responses to these sensations?

A. Glia C. Interneurons and motor neurons

B. Sensory neurons D. Neurons

11. Which cells in the central nervous system act as phagocytes, disposing of debris from dead cells?

glia

A. Astrocytes B. Oligodendrocytes C. Schwann cells D. Microglia

12. The presynaptic side of a synapse is usually

A. A dendrite B. A soma C. An axon terminal D. None of these

13. This glial cell is particularly important in maintaining homeostasis of the extracellular environment and is an important partner of neurons during synaptic activity. $[K^+]_o$

A. Astrocytes B. Oligodendrocytes C. Schwann cells D. Microglia

14. This type of glial cell myelinates axons in the brain and spinal cord COPS

A. Astrocytes B. Oligodendrocytes C. Schwann cells D. Microglia

15. At most synapses, information is said to be transformed from electrical to chemical to electrical. What is the "chemical"? NT

A. Synaptic vesicles C. Ion channels

B. Neurotransmitter AKRD D. membrane receptors

16. The motor molecules kinesin and dynein carry their cargo along microtubules. In which directions do they travel? A R

A. Either one can travel either in the retrograde or anterograde directions

B. Both kinesin and dynein carry cargo in the anterograde direction

C. Kinesin carries cargo in the anterograde direction, ~~dyne~~in in the retrograde direction

D. Kinesin carries cargo in the retrograde direction, dynein in the anterograde direction

→ continuous network of neurofiber

17. Golgi advocated the reticular theory, according to which

A. Neurons are separate cells that synapse upon each other, forming a reticulum → Cajal

B. Individual neurons are the elementary functional units of the brain

C. Neurites of different neurons are fused together to form a continuous network → Golgi

D. The cortex consists of radial units of interconnected neurons

18. The purpose of the trepanation procedure was likely to

A. provide a means to replace lost blood.

B. end the life of someone displaying symptoms of a mental disorder.

C. give evil spirits an "escape route."

D. cause headaches to overcome seizure symptoms.

19. During the period of the Renaissance, the prevailing theory regarding how muscles contract involved

A. electrical impulses from the brain traveling through nerves to muscles.

B. vibrations from the pineal gland pushing blood through gray matter.

C. fluid from the ventricles moving through hollow nerves and engorging muscles.

D. animal spirits from white matter moving through gray matter to muscles.

20. Galen (130-200 C.E.) suggested that the cerebrum of the brain must receive sensations and the cerebellum must command the muscles. What primary pieces of evidence led him to arrive at these conclusions?

A. The nerves of all muscles go from the spinal cord to the cerebellum.

B. The cerebrum was soft to the touch and the cerebellum firm.

C. Nerves from the eyes, ears, and tongue travel to the cerebellum.

D. Nerves from the heart travel to the cerebellum, and those from muscles to the cerebrum.

Part II : Free Response Questions 4*10=40 marks

Choose 4 of the following questions

Nissl	Golgi
distinguish neuron & glia	no distinguish
only label cell body/soma	all subcellular structure. soma. axon. dendrites
most cell could be colorized	only 3~5% been colorized.
lamellation//layers, Cytoarchitecture	details. dendritic spines

3
3
4

1. **Distinguish** between Nissl Stain Method and Golgi Stain Method.

2. Using **THREE** named examples of glia cell, **State** their specific functions.

Astrocyte: regulate extracellular NT. K^+ ions
microglia: act like phagocyte. immunity
oligodendroglia: mylenation cell in CNS. insulate.
ependymal cell: forming CSF, guid cell development

3. Structure features reflect functional roles, **state** the roles of dendrite, soma, dendritic spine and axon in nerve signaling.

signal input summation intergrate forming sympse. signal output

Galen: discover cerebrum. cerebellum. ventricle.

4. Using **THREE** historical neuroscientists that *we learned in Chapter 1 as examples*. **Stated** their **name** and their **major contribution** to our understanding of the nervous system

Flurens: proves functions of cerebrum & cerebellum

Hippocrate.

Broca/Fritsch & Hitzig/Gall: Cerebral localization

5. Based on the examples of Macaque and Mice, **evaluate** the statement: "*brain structure reflects the organism's adaptation on environment*"

Galvanin/Raymond: Nerve is wire. Magendie & Bell. unidirectional

SDMV

whisker

visual cortex / barrel cortex

6. State **THREE** examples of neural disorders, with the name and major symptom/features listed.

PD. AD. Autism, spinal paralysis.

7. State **THREE** differences in subcellular component of **axon hillock** and **axon terminal.**

MT	✓	✗
SV	✗	✓
Mitochondria	✗	✓

8. **State TWO** different methods of classifying neurons and list the possible types recognized.

Examples: with dendritic spine or not, spiny & aspinous.

- number of neurite, unipolar & bipolar & multipolar
- dendrite shape, stellate & pyramidal
- connection, sensory neuron. interneuron, motor neuron
- axon length, Golgi I & II.
- Gene Expression, Inhibitory & Excitatory.